Aprendizaje automático

Ciencia de Datos e Inteligencia Artificial - 3 º Curso
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Diego José García Callejas – dj.garcia.callejas@alumnos.upm.es

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01.

Introducción

0.1 Introducción

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CSV

Data-odd.csv

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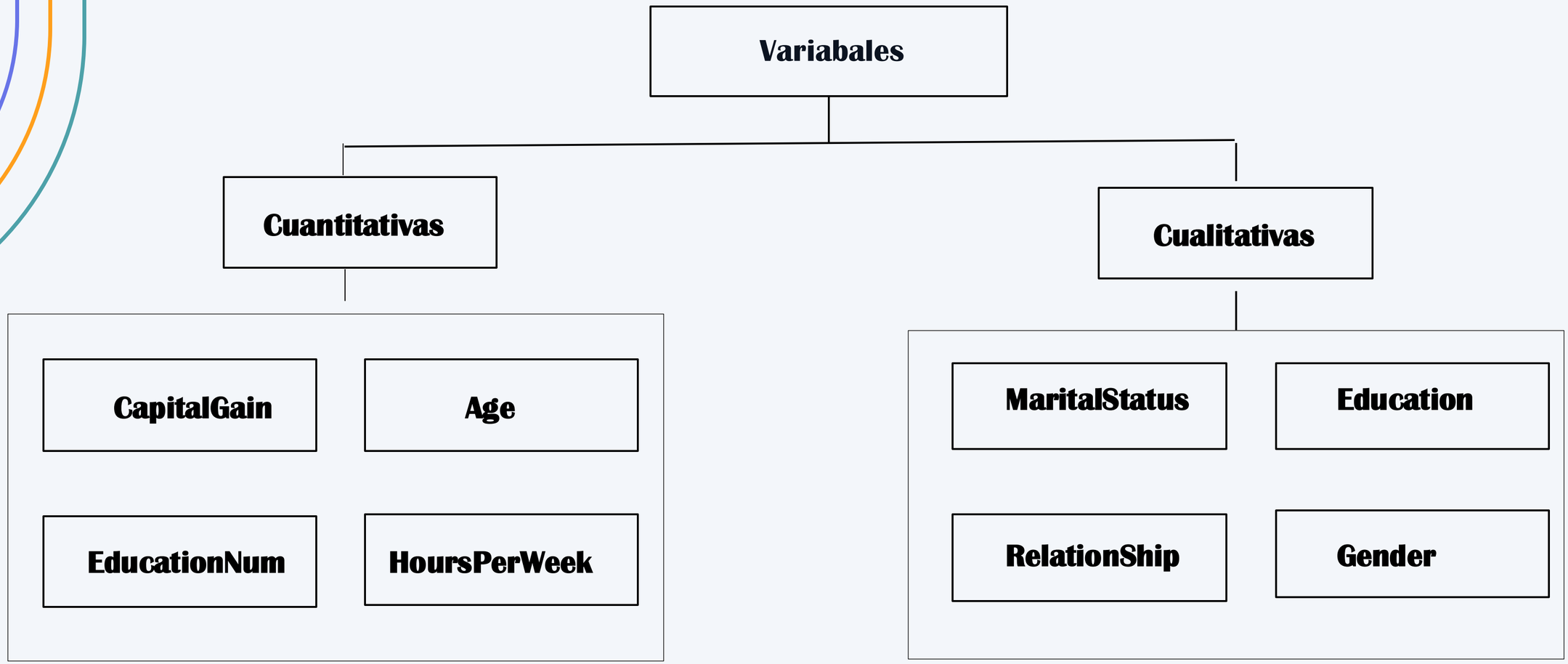
Análisis del mejor agrupamiento y relaciones

02.

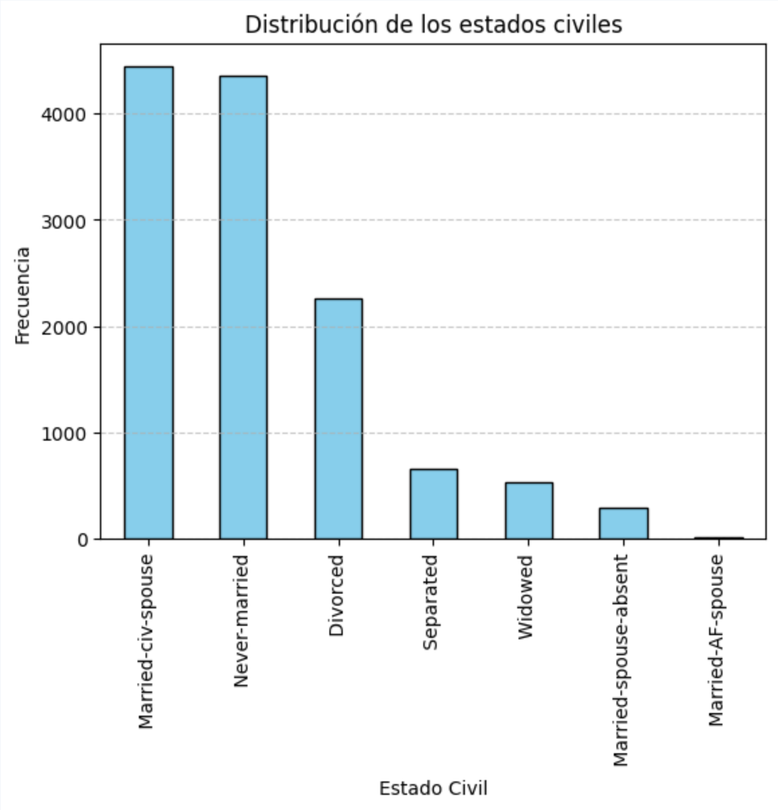
Análisis

02. Análisis

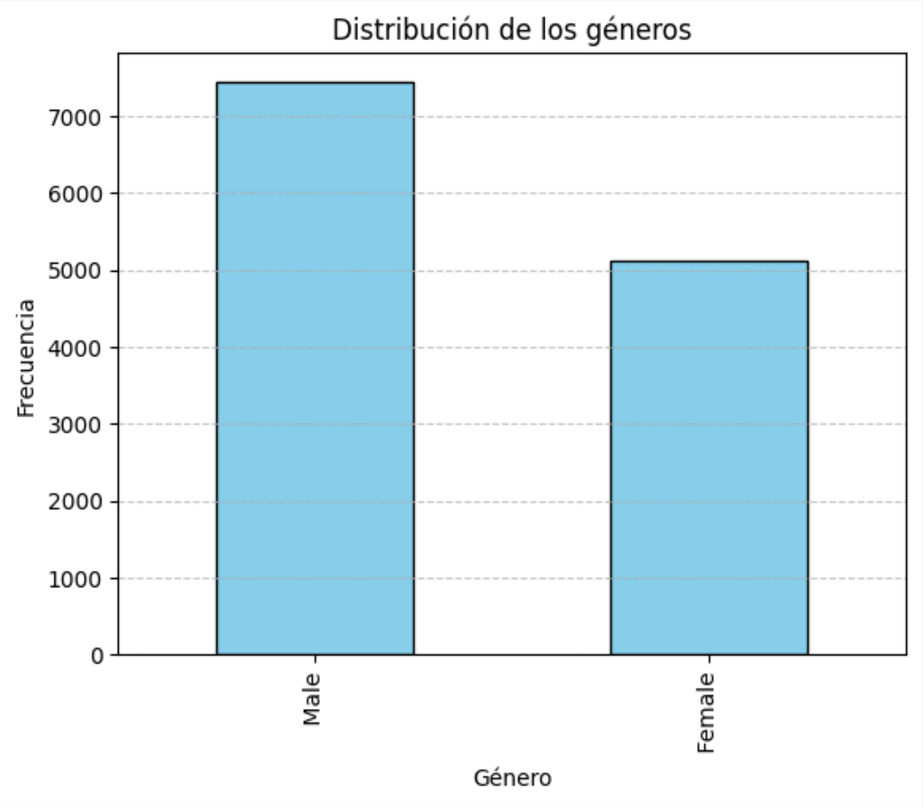

Variabales
Cuantitativas
Cualitativas
CapitalGain
Age
EducationNum
HoursPerWeek
MaritalStatus
Education
RelationShip
Gender

0.2 Análisis

MaritalStatus

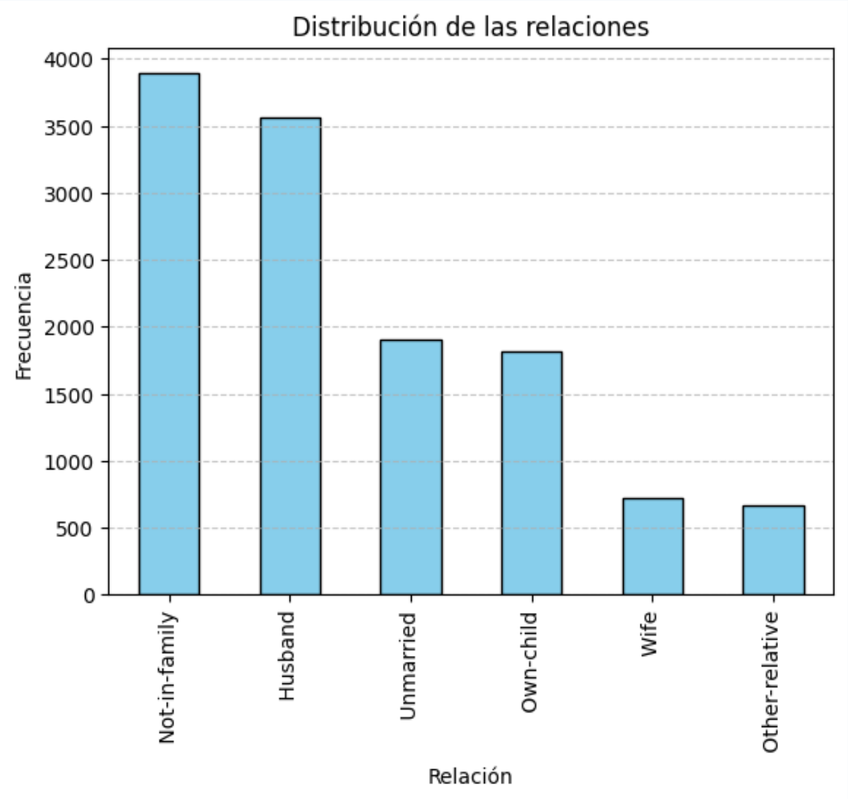


Gender

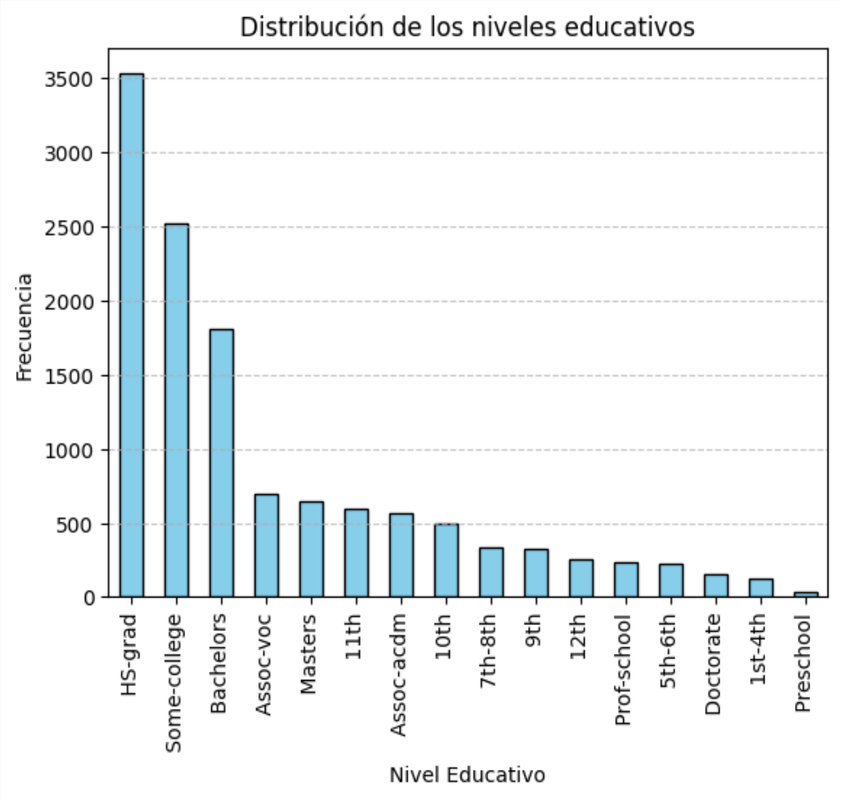


0.2 Análisis

Relationship

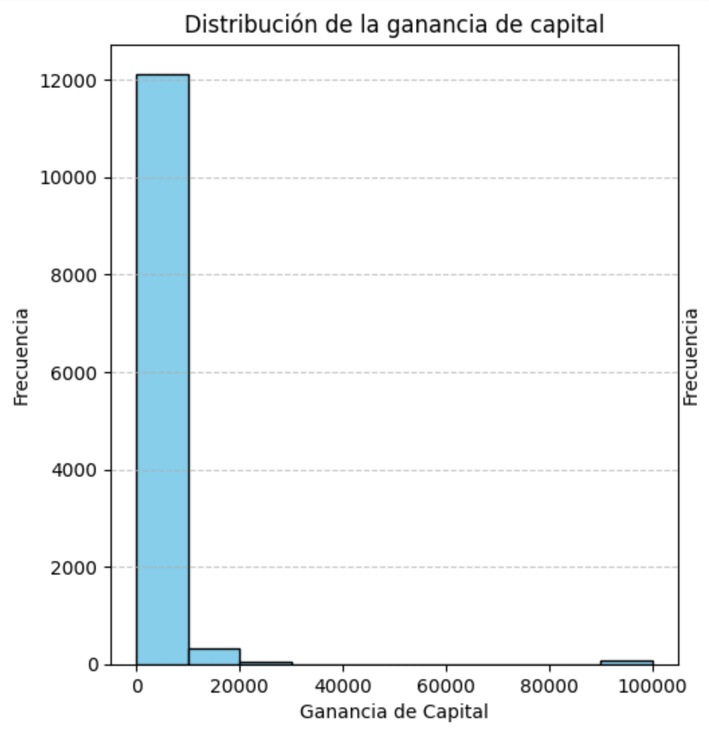


Education

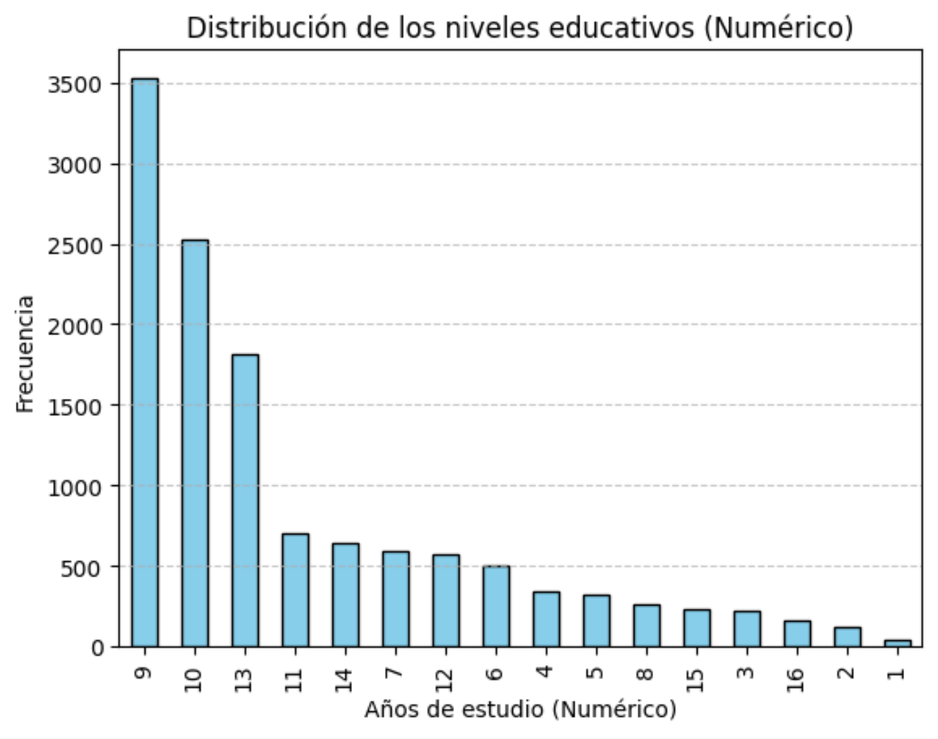


0.2 Análisis

CapitalGain

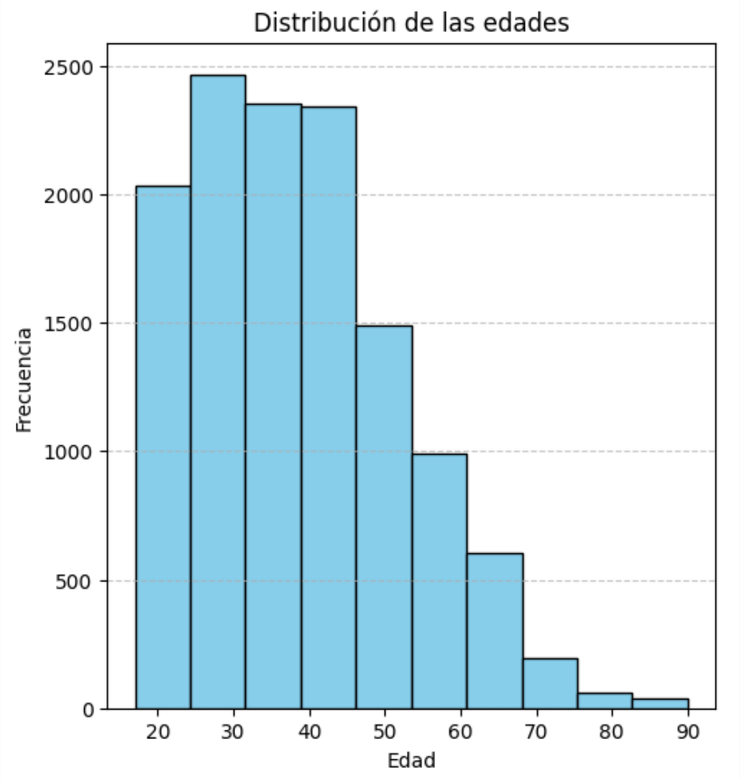


EducationNum

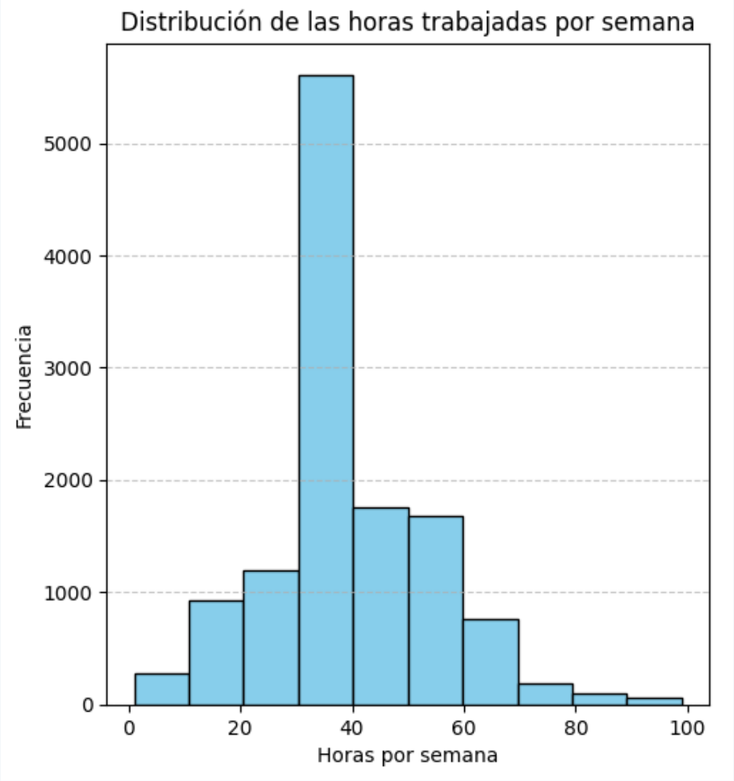


0.2 Análisis

Age



HoursPerWeek



03.

Escalado

03.Escalado

Variables Cuantitativas

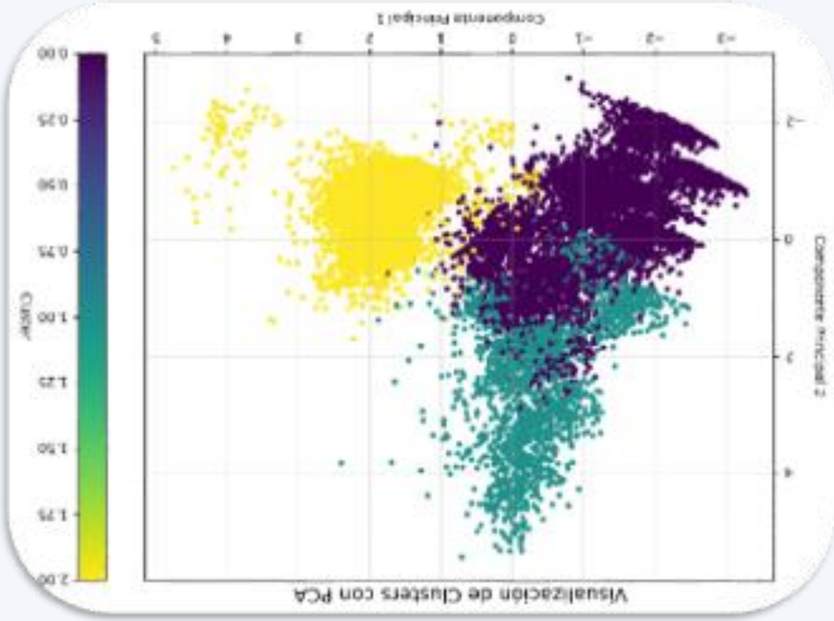
MaritalStatus
Relationship
Gender

OneHotEncoder

03.Escalado

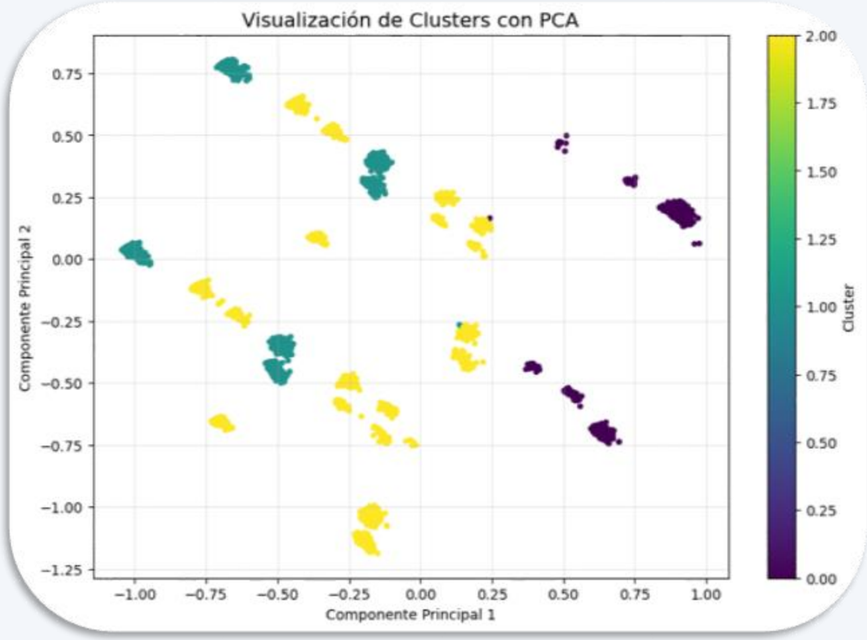
1

StandarScaler



2

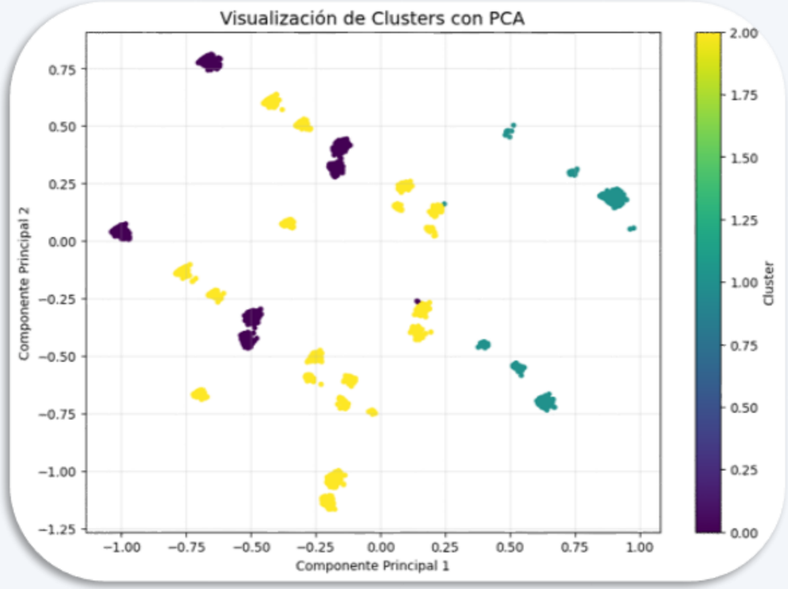
MinMaxAbs



03.Escalado

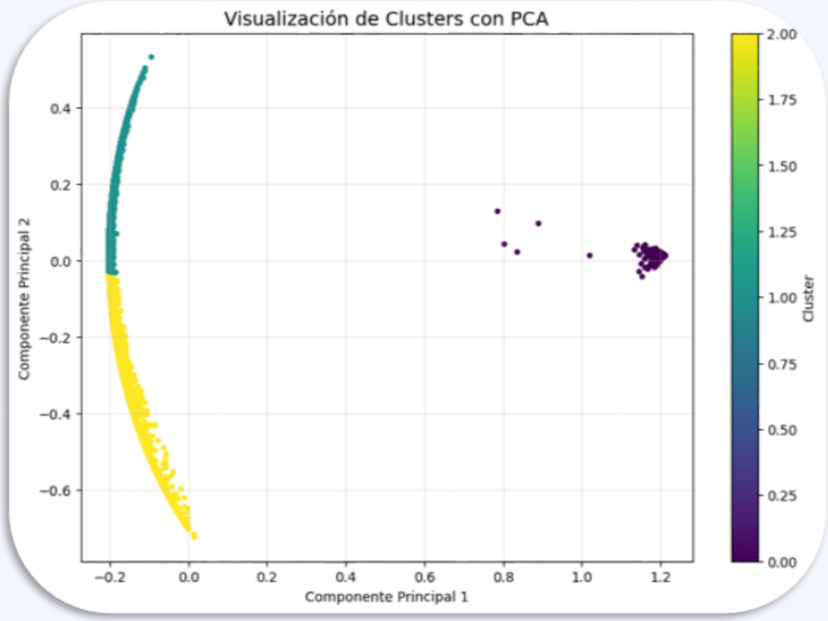
3

MaxAbsScaler

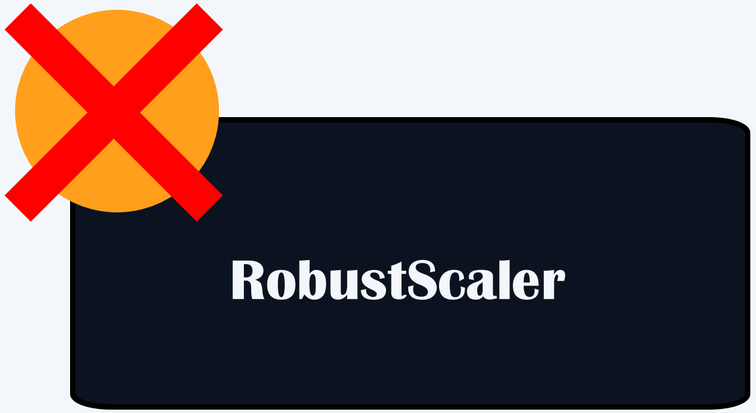


4

Normalizer



03.Escalado



03.Escalado

6

Personalizado

Variables OneHotEncoder

StandarScaler/nº var binarias

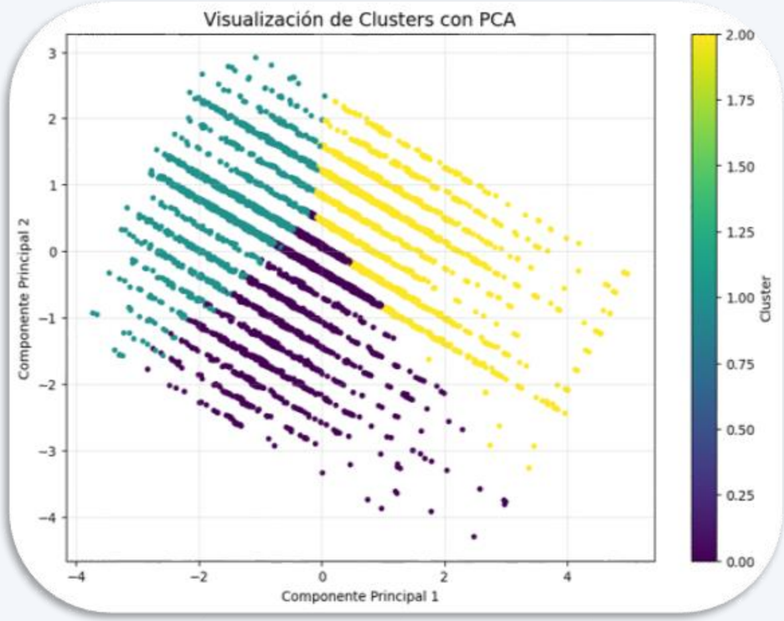
EducationNum
Age
HoursPerWeek

RobustScaler

CapitalGain

Valores atípicos y excaso

MaxAbsSclaer



04.

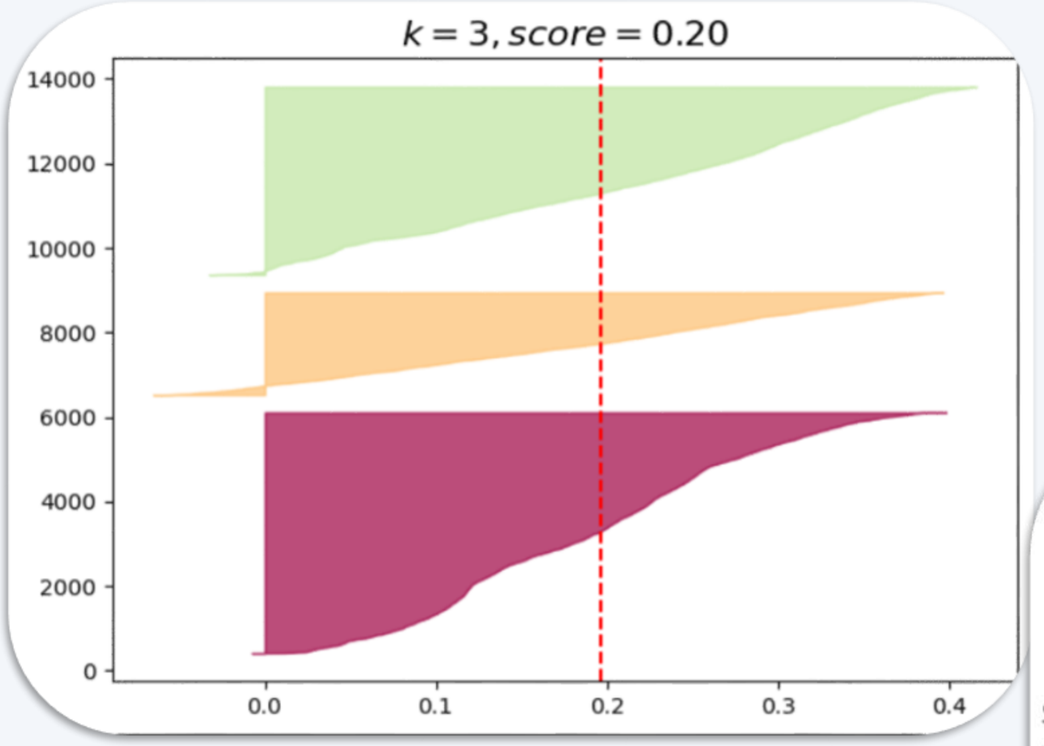
Clustering

04. Clustering

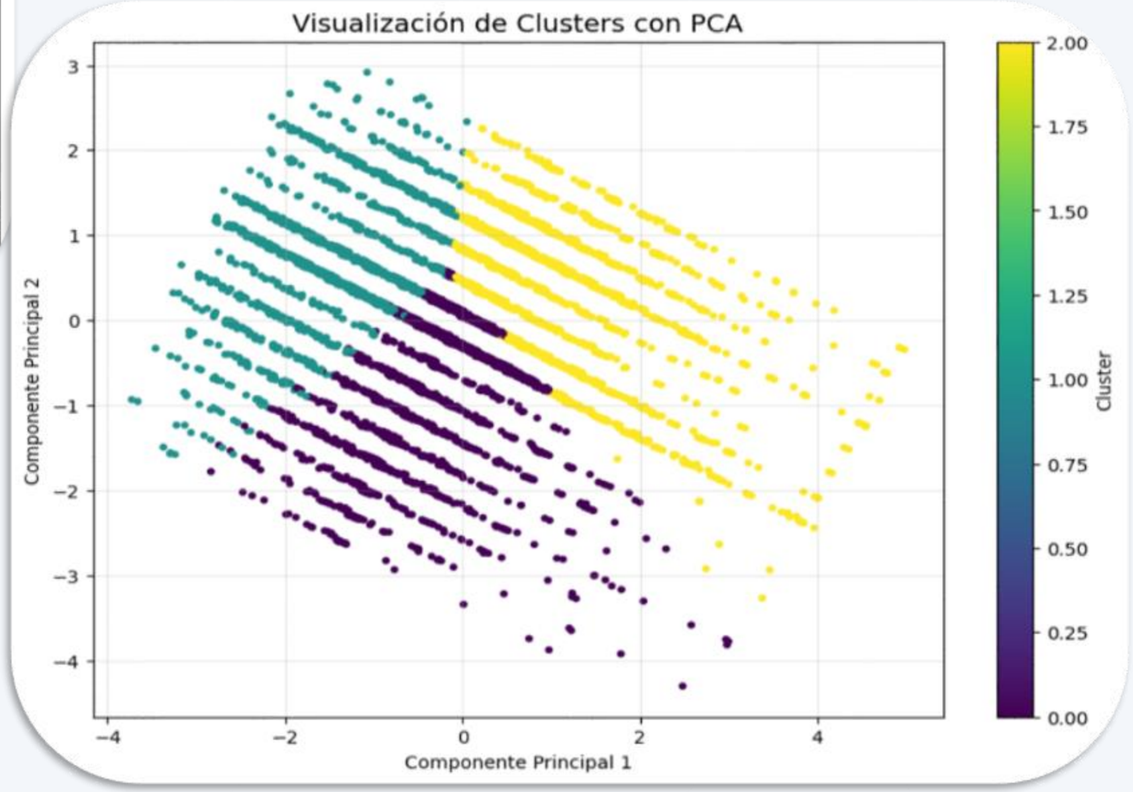
1. KMeans, k=3
2. KMeans, con distintos k
3. Jerárquico
4. DBSCAN
5. Mixturas de Gaussianas

KMeans, K=3

Silhouette



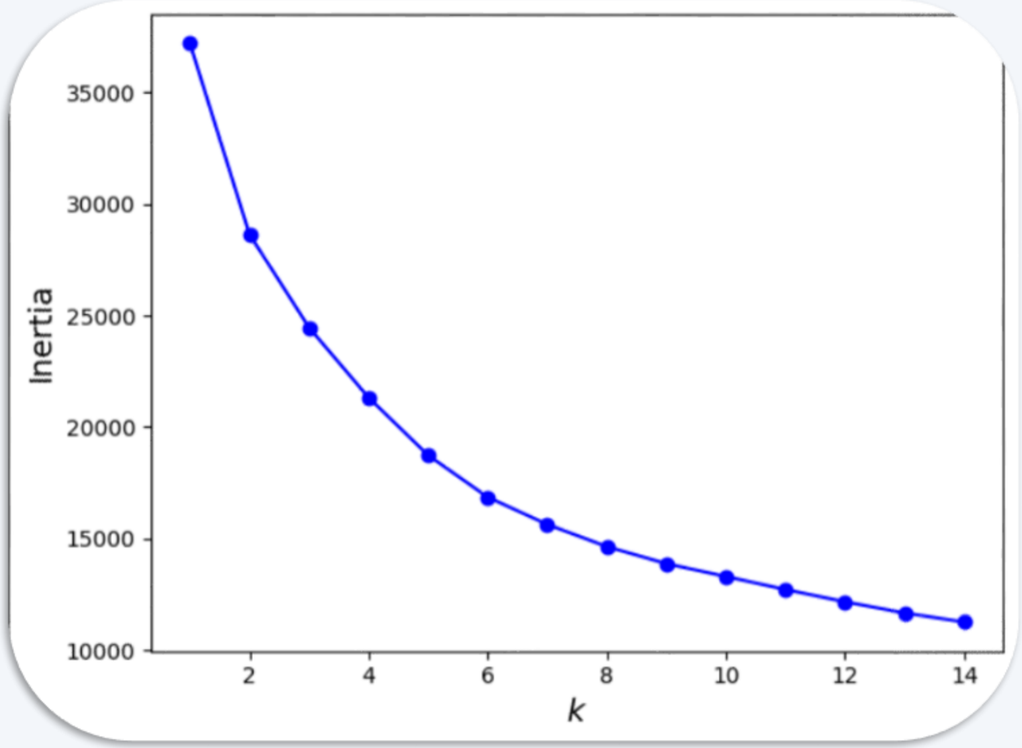
PCA



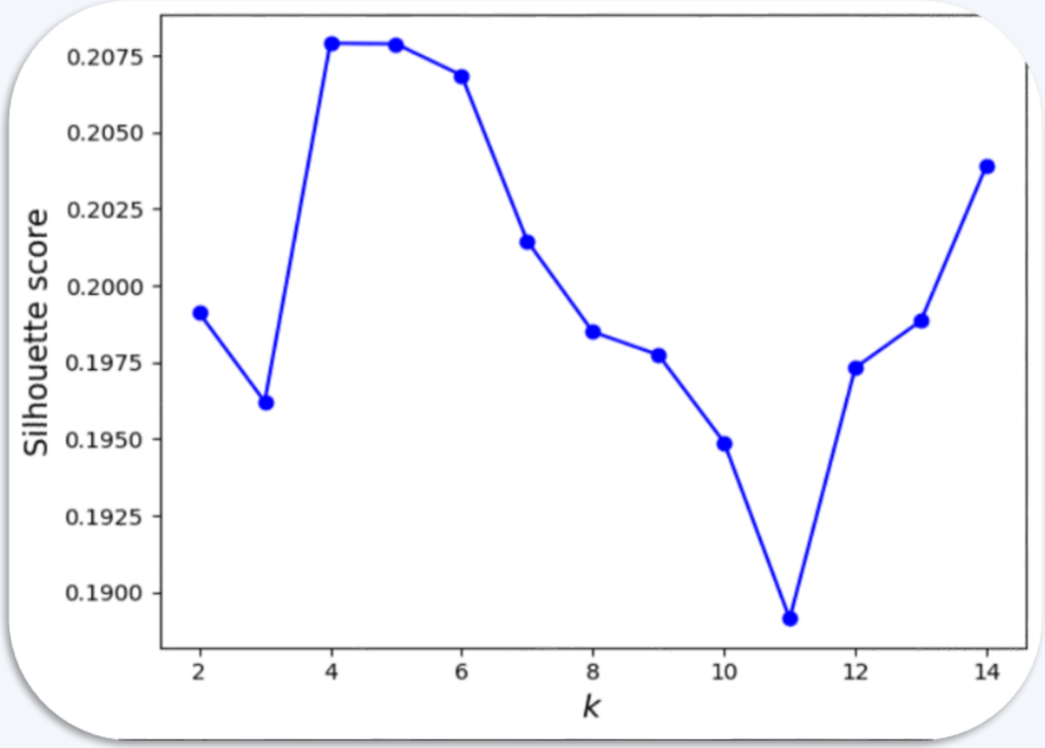
KMeans, distinto K.

Selección de K

Inercia



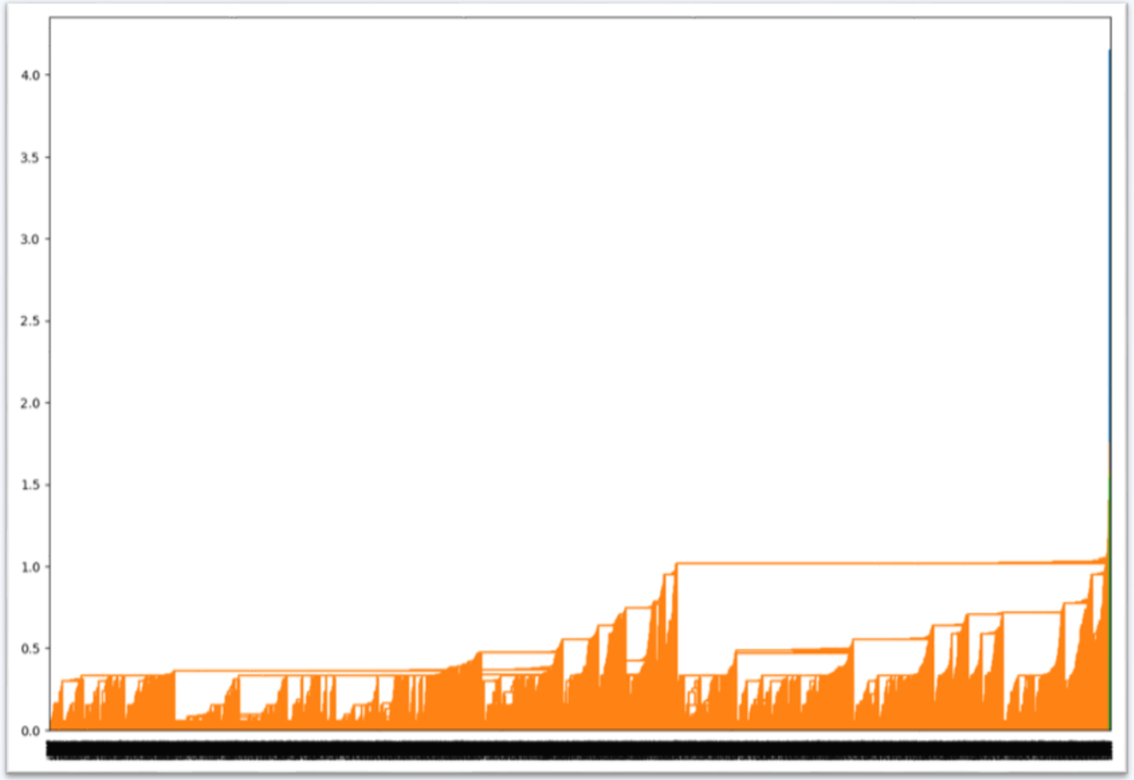
Silhouette



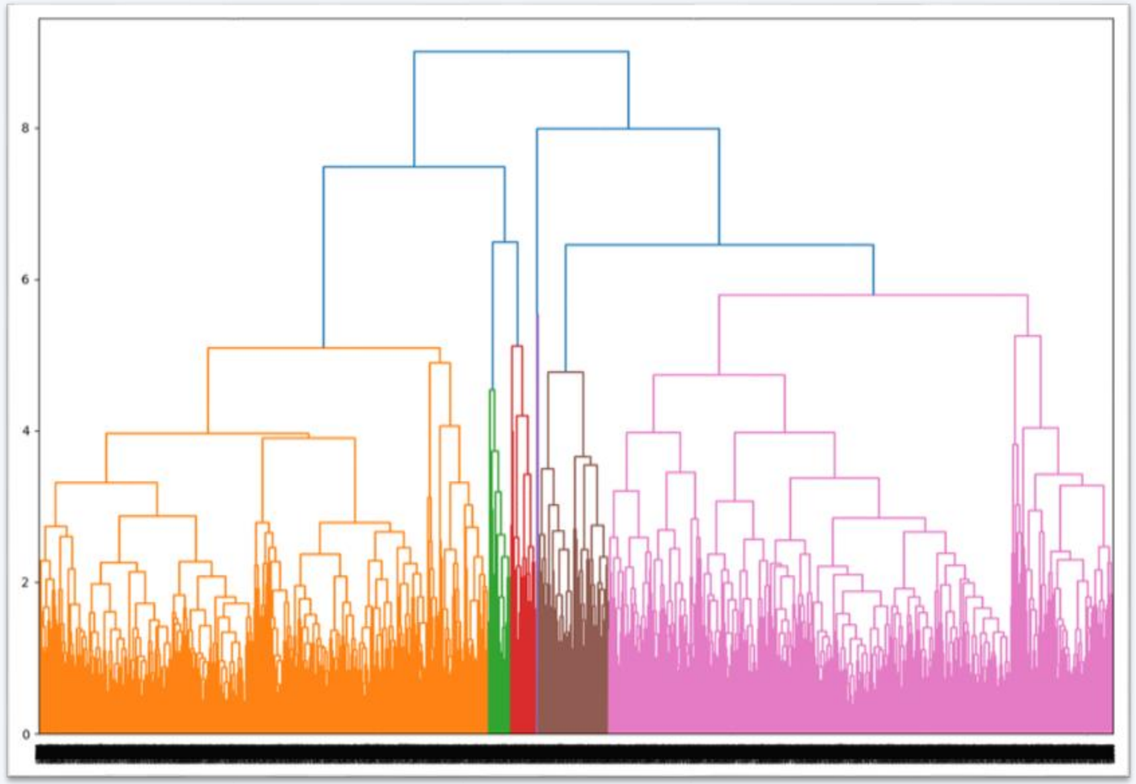
Jerárquico.

Single y Complete

Single



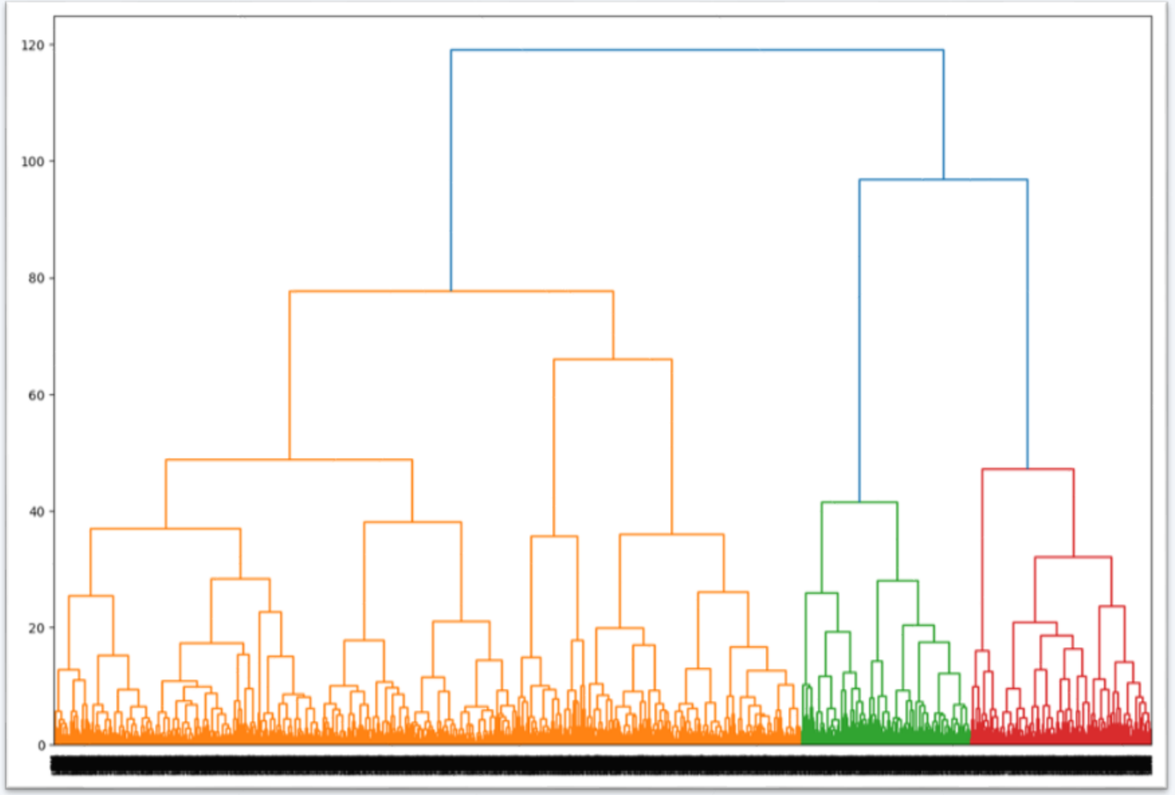
Complete



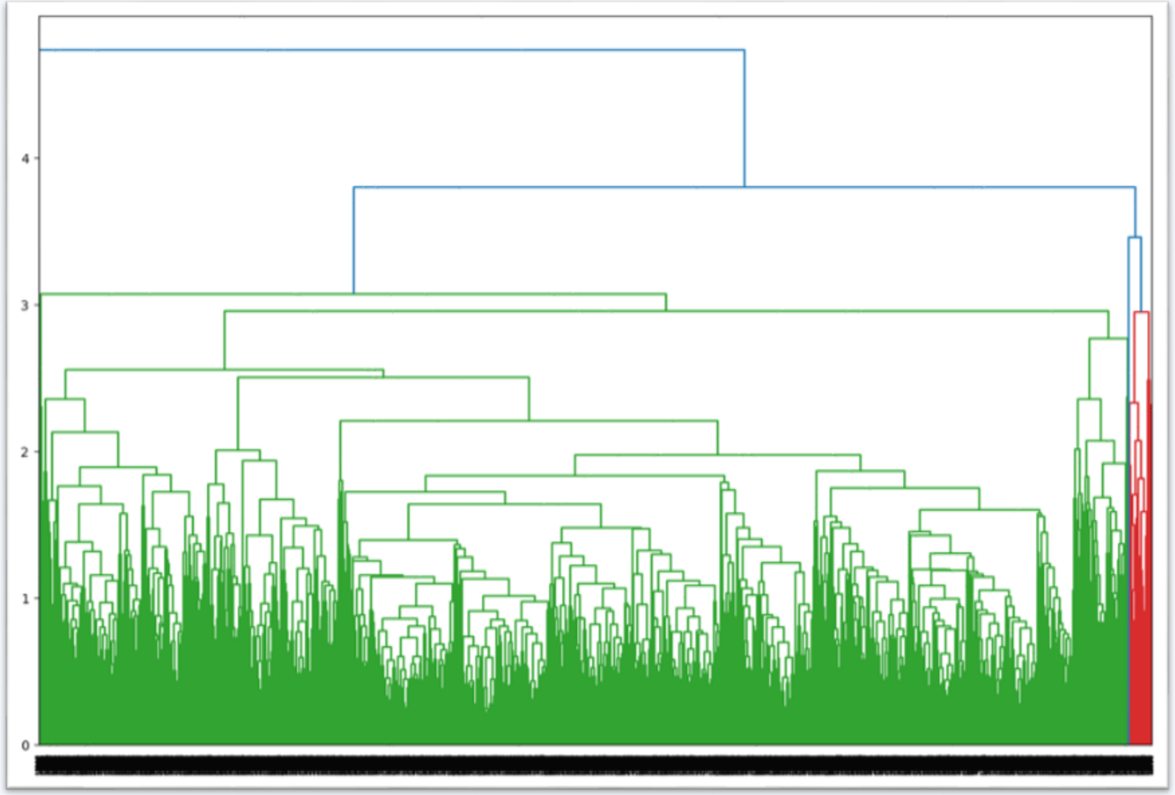
Jerárquico.

Ward y Average

Ward

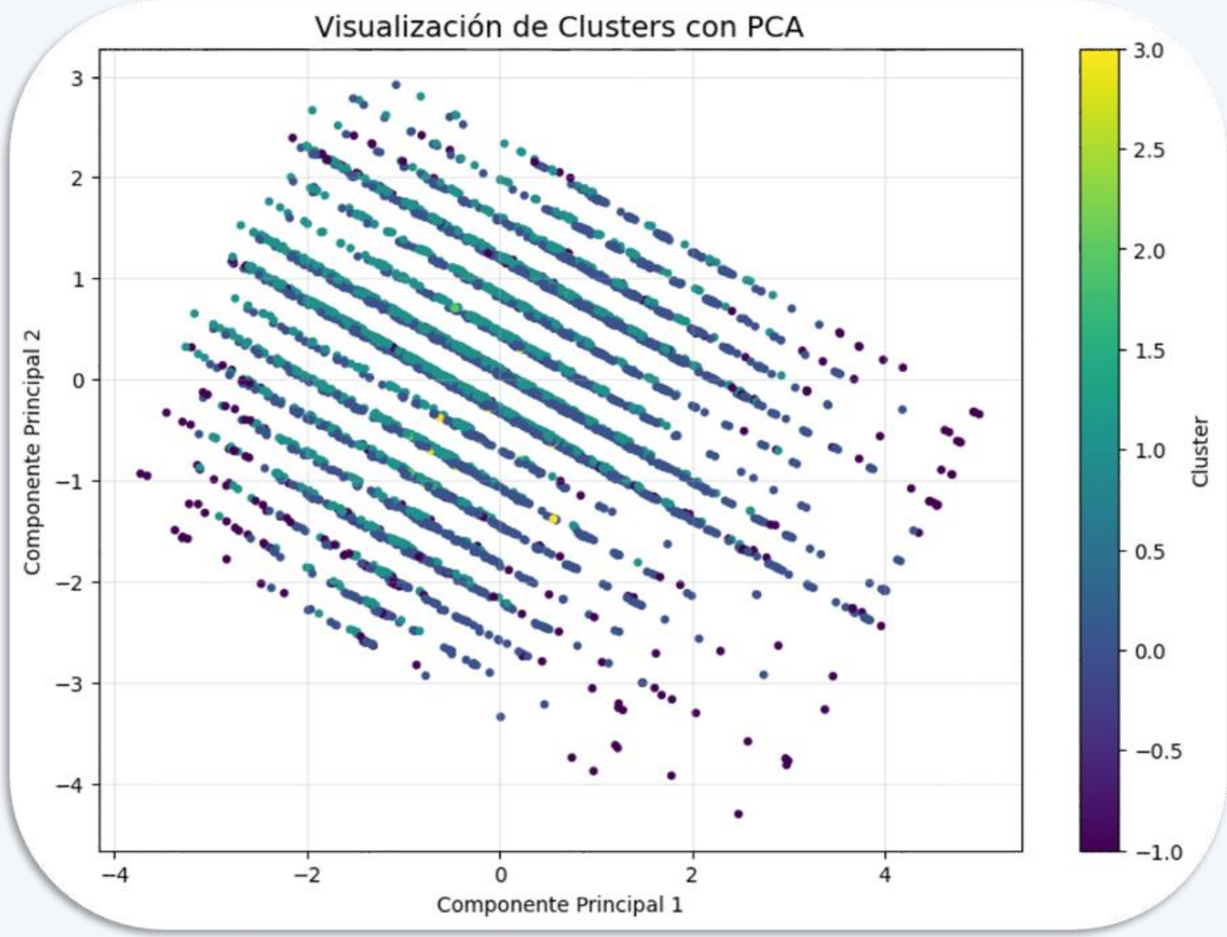


Average



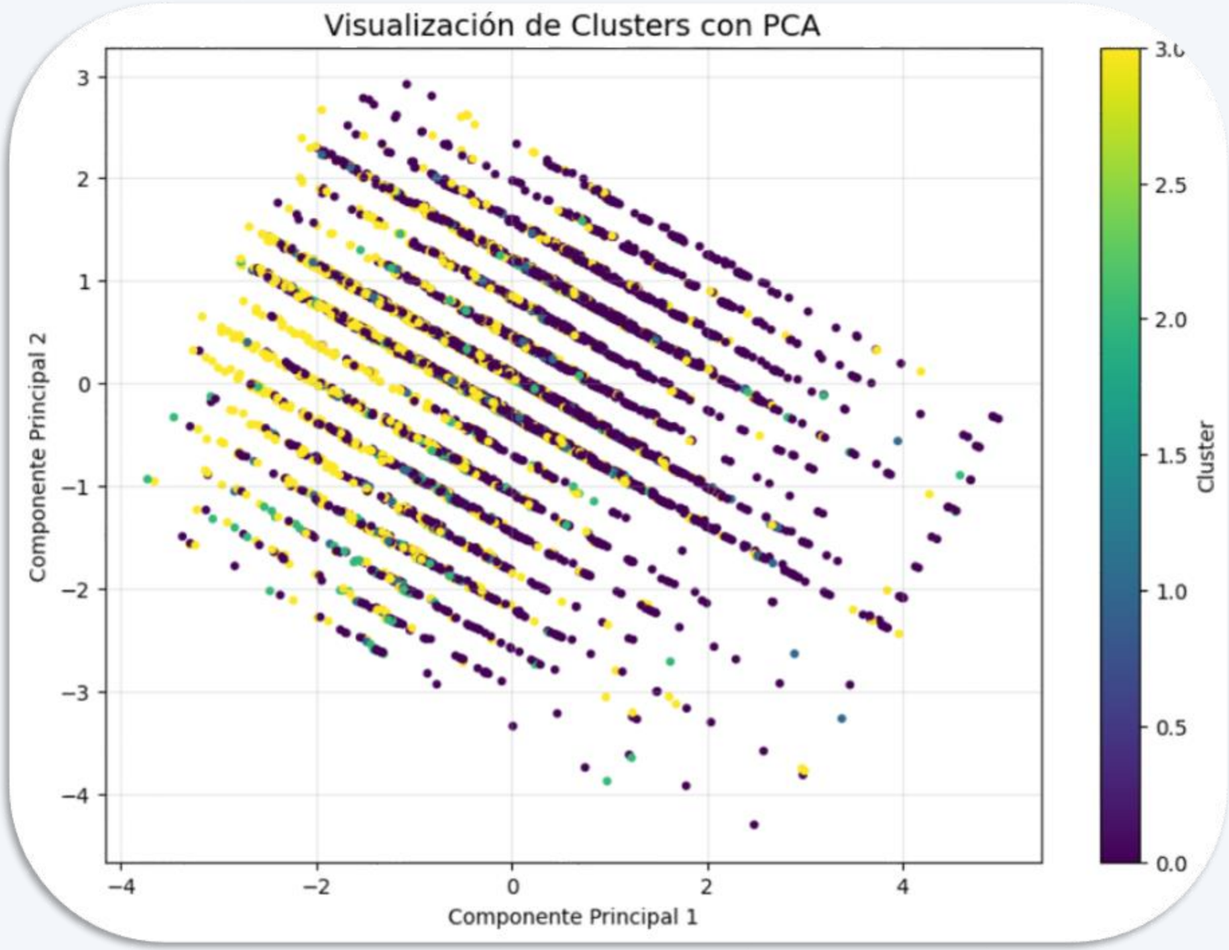
DBSCAN

PCA


Visualización de Clusters con PCA
Componente Principal 2
Componente Principal 1
Cluster
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0

Mixturas de Gaussianas

PCA

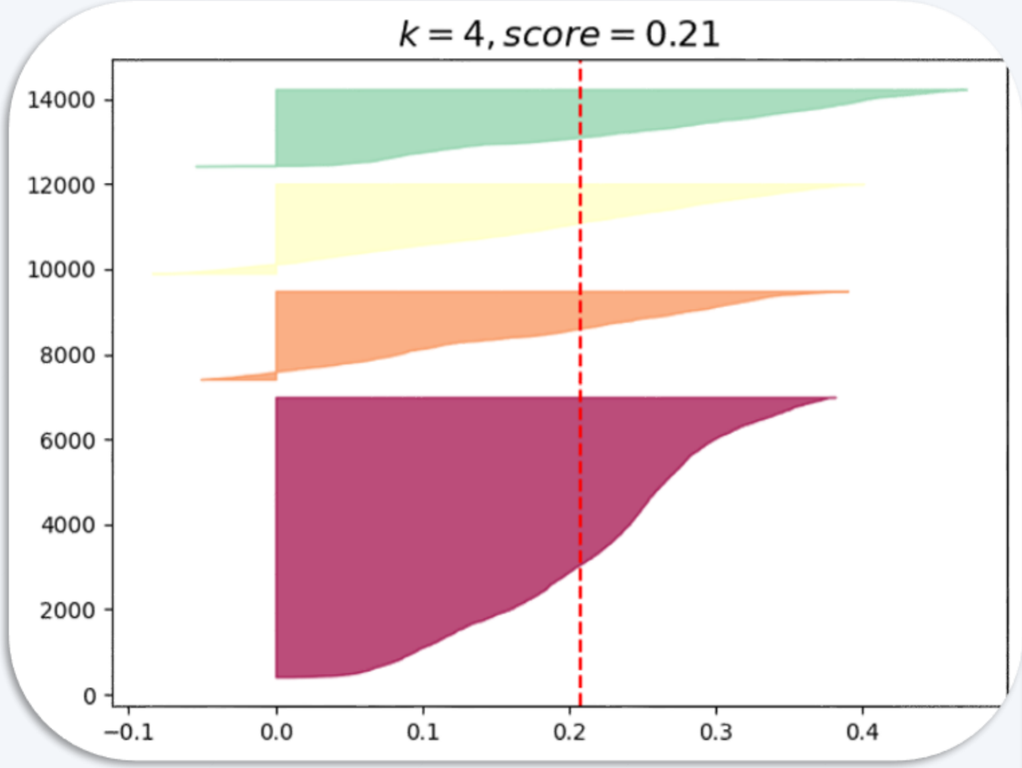


05.

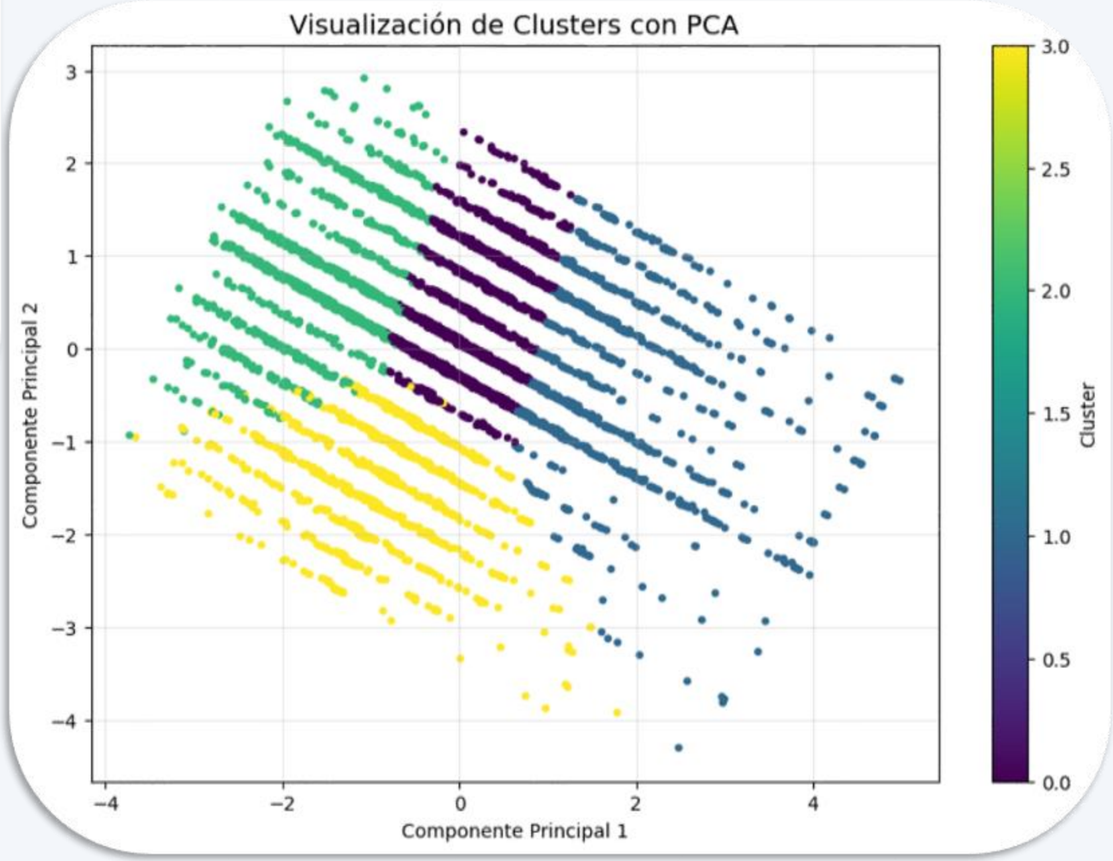
Conclusiones

Análisis del mejor método

Silhouette

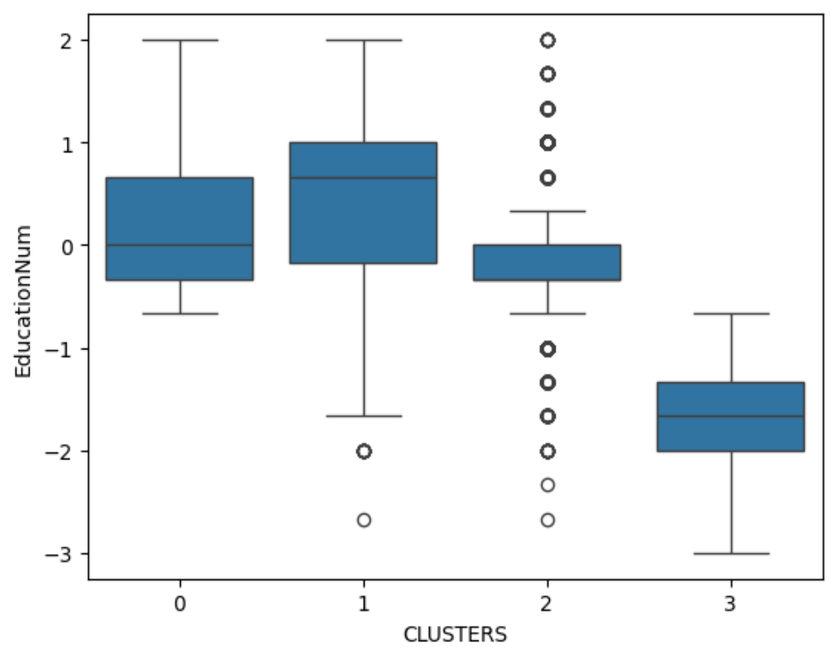


PCA

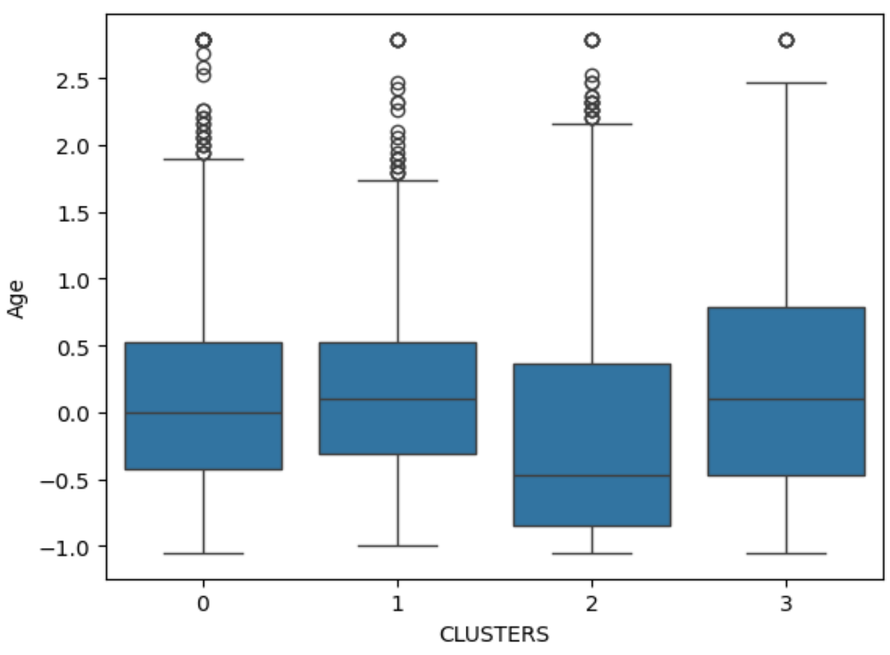


Relaciones

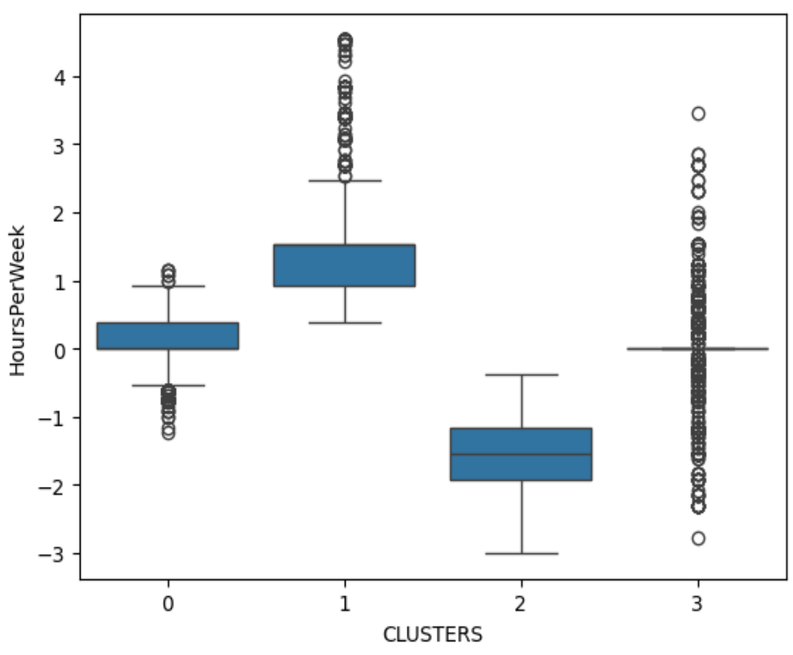
Education num



Age



HoursPerWeek



¡¡Muchas gracias!!